

PRODUCT REQUIREMENTS DOCUMENT

AquaFlow B2B

Water Delivery Management System

Admin Dashboard & Customer Portal

Version 1.0

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Prepared for: Client Review

Status: Draft for Discussion

1. Project Overview & Goals

This document defines the requirements, features, and user flows for AquaFlow B2B, the Water Delivery Management System, so the client and the development team are aligned on what will be built and why.

AquaFlow B2B is a two-sided platform that connects customers who need water delivered with an operations team that manages supply, fleet, and billing.

1.1 Goals

- Give the operations/admin team a single dashboard to monitor demand, capacity, deliveries, and revenue.
- Let customers self-serve: request water, track status, and pay invoices without calling support.
- Reduce manual work in request approval, fleet assignment, and invoicing.
- Provide the business with real-time, reliable analytics to plan capacity and revenue.

2. User Roles

Role	Description
Admin / Operations	Reviews and approves requests, assigns fleet, generates invoices, and monitors business KPIs.
Customer	Submits water requests, tracks orders and deliveries, and pays invoices online.

3. User Flows

3.1 Admin Flow

1. View KPIs on the dashboard — Daily Capacity, Pending Requests, Active Deliveries, Revenue.
2. Monitor analytics — Revenue & Volume Trend, Regional Capacity, Customer Utilization.
3. Manage incoming water requests.
4. Approve or reject each request.
5. On approval, an order is created.
6. Assign fleet and schedule the delivery.
7. Generate the invoice for the order.
8. Customer completes payment.
9. Order is marked complete.

3.2 Customer Flow

10. Create a water request.
11. Track the request status (pending / approved / rejected).
12. View the resulting order once approved.
13. Track the delivery in progress.
14. View and pay the invoice.
15. Order marked as completed.

3.3 End-to-End Process Flow

The two flows above meet at a shared process. This is the backbone the whole AquaFlow B2B system is built around:

Step	Owner	Outcome
1. Customer submits water request	Customer	Request enters admin queue
2. Admin reviews request	Admin	Request evaluated against capacity
3. Approve / Reject	Admin	Customer notified of decision
4. Order created	System	Order record generated on approval
5. Delivery assigned	Admin	Fleet and route scheduled
6. Invoice generated	System	Billing document created
7. Customer pays	Customer	Payment recorded against invoice
8. Order completed	System	Order closed, history updated

4. Current Feature Set

4.1 Admin Dashboard

- KPI cards: Daily Capacity, Pending Requests, Active Deliveries, Revenue.
- Analytics: Revenue & Volume Trend, Regional Capacity, Customer Utilization.
- Request management with approve/reject actions.
- Fleet and delivery assignment.
- Invoice generation.

4.2 Customer Portal

- Submit a new water request.
- Track request status.
- View orders and delivery progress.
- View and pay invoices online.

5. Proposed Enhancements

Note on this section: The enhancements listed below are recommendations proposed by the development team, not requirements requested by the client. They are shared for consideration and are not yet approved for build. Any item the client wishes to proceed with should be confirmed and moved into the agreed AquaFlow B2B project scope.

The following improvements are grouped by area. These are recommended additions on top of the current system to improve usability, performance, and business insight.

5.1 Dashboard

- Click-through (drill-down) on KPI cards — e.g. clicking Pending Requests opens that filtered list directly.

- Show “last updated” time next to metrics so users know how fresh the data is.
- Customizable dashboard — show, hide, or rearrange widgets.
- Date range filters — Today, This Week, This Month, Custom — instead of today-only data.

5.2 Charts & Analytics

- Interactive charts with tooltips, zoom, and click-to-filter.
- Toggle between daily, weekly, monthly, and yearly trend views.
- Period-over-period comparison, e.g. this month vs. last month.

5.3 User Experience

- Loading skeletons while dashboard data loads.
- Clear, friendly empty states when there is no data to show.
- Clear error messages if a dashboard API call fails.
- Toast notifications and confirmation dialogs for key actions.

5.4 Navigation

- Clickable summary cards that redirect to the relevant filtered page.
- Breadcrumb navigation on inner pages.

5.5 Tables & Data

- Column sorting.
- Saved filters and improved search.
- Customizable visible columns.
- Bulk actions where applicable (e.g. bulk approve).
- Server-side pagination for performance on large datasets.

5.6 Order Tracking

- A visual order tracking timeline for customers, from request to completion.

5.7 Performance

- Fetch independent dashboard widgets in parallel rather than sequentially.
- Cache analytics data to reduce repeated load time.
- Real-time updates for dashboard and order statuses, without manual refresh.

5.8 Notifications

- Activity feed / notification panel for approvals, deliveries, and invoices.
- Notification filters: All, Unread, Warning, Payment, Delivery.
- Search within notifications and set notification preferences.
- Real-time notification delivery, without page refresh.

5.9 Export & Reporting

- Export dashboard reports and analytics to Excel, CSV, or PDF.

5.10 Accessibility & Consistency

- Consistent UI patterns, spacing, and components across all screens.
- Improved accessibility — keyboard navigation, color contrast, screen-reader support.

6. Functional Requirements Summary

ID	Requirement	Priority	Source
FR-1	Admin can view live KPIs and analytics on a central dashboard	High	Client
FR-2	Admin can approve or reject customer water requests	High	Client
FR-3	System auto-creates an order on request approval	High	Client
FR-4	Admin can assign fleet and schedule deliveries	High	Client
FR-5	System can generate invoices tied to an order	High	Client
FR-6	Customer can pay invoices online	High	Client
FR-7	Customer can track request, order, and delivery status	High	Client
FR-8	Dashboard cards support drill-down navigation with filters	Medium	Proposed
FR-9	Analytics support custom date ranges and period comparison	Medium	Proposed
FR-10	Tables support sorting, filtering, and bulk actions	Medium	Proposed
FR-11	System sends real-time notifications for key events	Medium	Proposed
FR-12	Reports/analytics can be exported to Excel, CSV, or PDF	Medium	Proposed
FR-13	Dashboard widgets are customizable and load in parallel	Low	Proposed

“Client” indicates requirements captured from the original project scope. “Proposed” indicates additions recommended by the development team, pending client approval.

7. Assumptions & Constraints

- Payment gateway integration is available and supports online invoice payments.
- Fleet and delivery data can be captured and updated by the admin team in real time.
- Export functionality (Excel/CSV/PDF) will use existing analytics data sources.
- Real-time updates depend on a supporting notification/websocket infrastructure being in place.